Task 17: Multi-VPC AWS network environment

Objective

Design and implement a **Multi-VPC AWS network environment** across **two regions** with **advanced subnetting**, and establish connectivity using **VPC Peering**.

Problem Statement

ABC Technologies is expanding its AWS infrastructure across multiple business locations and wants to build a more structured inter-office network architecture in AWS.

The company now has **multiple HQ environments** and **multiple branch offices**. The requirement is to host the **HQ environments in** `us-east-1` and all **branch offices in** `us-east-2`.

Instead of using a single VPC, the company wants to create **separate VPCs for each HQ and each office**, and connect them using **VPC Peering**. This is intended to help the team understand **multi-VPC design, multi-region connectivity, advanced subnet planning, and controlled communication between networks**.

Each VPC should contain both:

- **Public subnets**
- **Private subnets**

These subnets should be designed with clear use cases. For example:

- **Public subnets** for bastion hosts, public-facing applications, or NAT gateways
- **Private subnets** for internal application servers, databases, or backend services

As a DevOps intern, your task is to design the network, create the VPCs in AWS, perform subnetting based on host requirements, connect the VPCs using peering, deploy EC2 instances for testing, and verify connectivity.

Network Requirements

Region 1: HQ Environments

Location: `us-east-1`

Create **2 HQ VPCs**:

HQ-1

- Public Subnet 1 → **100 hosts**
- Private Subnet 1 → **200 hosts**
- Private Subnet 2 → **50 hosts**

HQ-2

- Public Subnet 1 → **50 hosts**
- Private Subnet 1 → **100 hosts**
- Private Subnet 2 → **50 hosts**

Region 2: Branch Offices

Location: `us-east-2`

Create **4 Office VPCs**:

Office A

- Public Subnet → **50 hosts**
- Private Subnet → **100 hosts**

Office B

- Public Subnet → **30 hosts**
- Private Subnet → **50 hosts**

Office C

- Public Subnet → **50 hosts**
- Private Subnet → **100 hosts**

Office D

- Public Subnet → **30 hosts**
- Private Subnet → **50 hosts**

Task Requirements

1] Network Design

Create a network design for:

- **2 HQ VPCs in** `us-east-1`
- **4 Office VPCs in** `us-east-2`

Ensure:

- No overlapping CIDR ranges between any VPCs or subnets
- Proper subnetting based on host requirements
- Clear distinction between public and private subnets
- Logical use of public and private subnet design
- Route planning is done correctly for inter-VPC communication

Students should decide the final CIDR ranges themselves.

2] AWS VPC Environment Setup

Create the required VPCs in AWS:

In `us-east-1`

- HQ-1 VPC
- HQ-2 VPC

In `us-east-2`

- Office A VPC
- Office B VPC
- Office C VPC
- Office D VPC

Each VPC must include:

- At least **1 public subnet**
- At least **1 private subnet**
- Internet Gateway where required
- Appropriate route tables

For private subnet use cases, students may include:

- Internal application servers
- Database servers
- Backend services

For public subnet use cases, students may include:

- Bastion host
- Public web server

- NAT gateway

3] VPC Peering

Set up **cross-region VPC peering** between:

- HQ-1 ↔ Office A
- HQ-1 ↔ Office B
- HQ-2 ↔ Office C
- HQ-2 ↔ Office D

Also set up **intra-region peering** between:

- HQ-1 ↔ HQ-2
- Office A ↔ Office B
- Office C ↔ Office D

Update route tables appropriately so communication works only where required.

4] EC2 Deployment and Connectivity Testing

Launch EC2 instances in:

- Public subnet of each HQ VPC
- Private subnet of each HQ VPC
- Public subnet of each Office VPC
- Private subnet of each Office VPC

Configure security groups to allow testing.

Verify connectivity between required VPCs using basic commands such as:

```
telnet <private-ip> 22

```

or equivalent connectivity checks.

Also validate:

- Public subnet instances have internet access where applicable
- Private subnet instances follow intended routing paths
- Peered VPCs can communicate based on route table design

5] Documentation

Document:

- Chosen VPC CIDR ranges
- Subnet CIDR ranges and host allocation
- Public and private subnet use cases
- Region-wise VPC setup
- Peering configuration
- Route table updates
- EC2 placement
- Connectivity test results

Expected Deliverables

Submit the following:

- Network design diagram showing all HQ and Office VPCs
- Screenshot of VPC peering and route table configuration
- EC2 deployment and connectivity proof
- Requirement analysis (Confluence link)
- Technical documentation (Confluence link)
  - Prepare a short document explaining: (Confluence link)
    - Public subnet vs Private subnet
    - VPC Peering vs Transit Gateway
    - Benefits of multi-VPC architecture
    - Also include a short video (2–3 mins) explaining the same documentation

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Time to Complete

3 Hours